Basic notions of knot theory

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Def A knot in $\mathbb{R}^3$ or in S^3 is a smooth embedding $K: S^1 \hookrightarrow \mathbb{R}^3$ (or S^3) w/ orientation

$L: S^1 \sqcup \cdots \sqcup S^1$ (n oriented) (link)

as well as the image of embeddings.

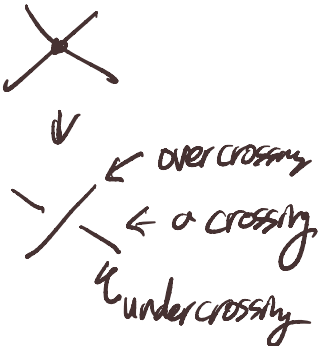


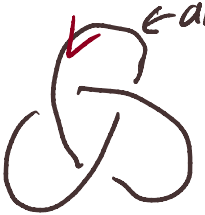
Def Two knots (or links) K_1 and K_2 are equivalent, $K_1 \sim K_2$ (oriented)

if $\exists H: \mathbb{R}^3 \times [0,1] \to \mathbb{R}^3$ s.t. ① $H|_{\mathbb{R}^3 \times \{t\}}: \mathbb{R}^3 \times \{t\} \to \mathbb{R}^3$ · orientation preserving homeo.

② $H|_{\mathbb{R}^3 \times \{0\}} = id_{\mathbb{R}^3}$, $H|_{\mathbb{R}^3 \times \{1\}}(K_1) = K_2$





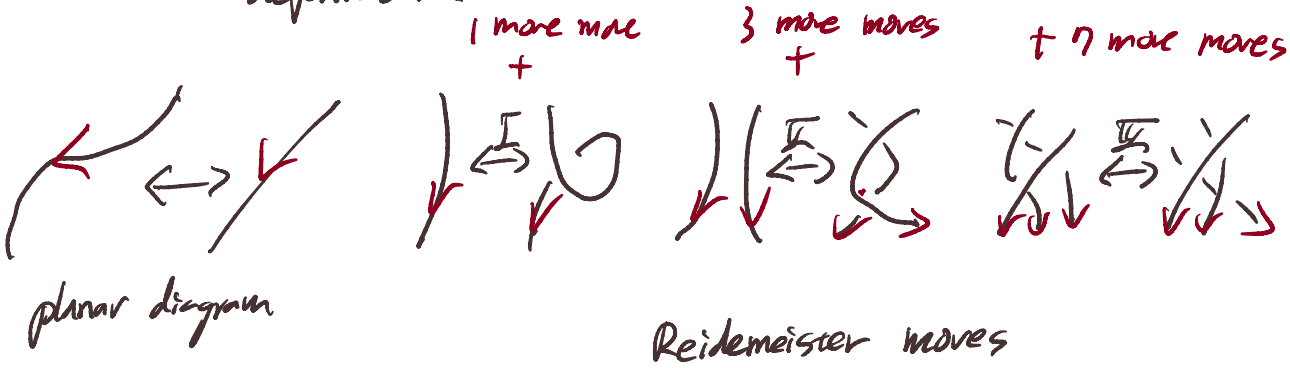


link : a (knot) diagram D_K

Thm (1927, Reidemeister)

K_1, K_2: links, D_{K_1}, D_{K_2}: diagrams of K_1 and K_2, resp.

$K_1 \sim K_2$ iff D_{K_2} can be obtained from D_{K_1} by a finite seq. of the following local deformations:

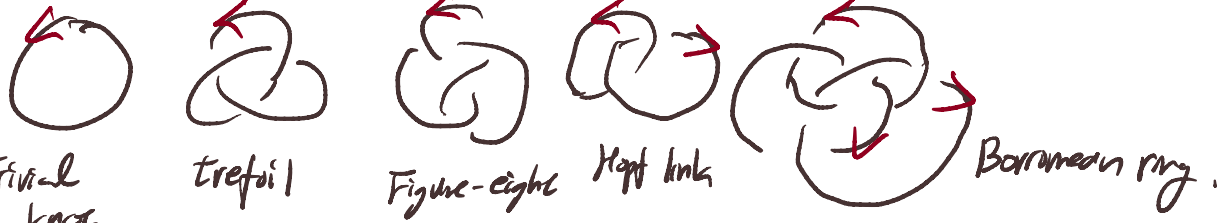


planar diagram

Reidemeister moves

Remark Thanks to Reidemeister theorem, "topological" object is changed to "combinatoric" object.

Ex.



Trivial knot

Trefoil

Figure-eight knot

Hopf link

Borromean ring.

Def (crossing sign)



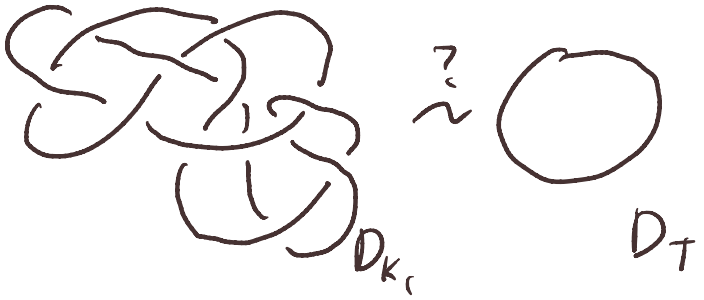
+1 −1 ← crossing sign of c



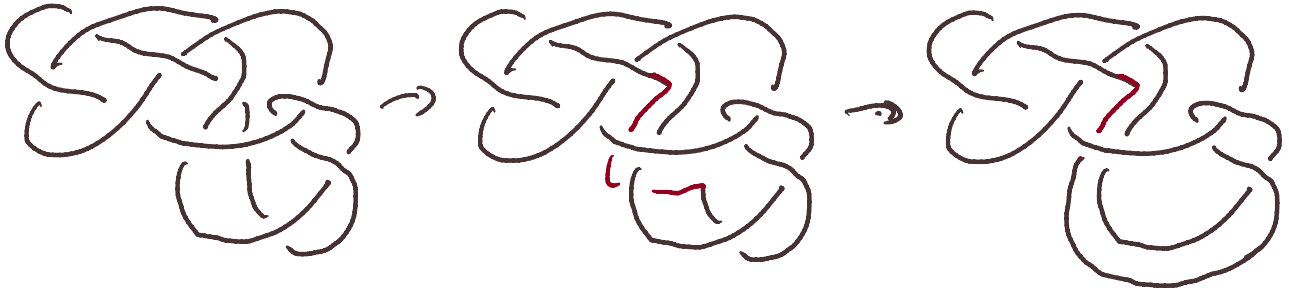
(v_1, v_2)

Main goal To distinguish knots and links in S^3.

Exa



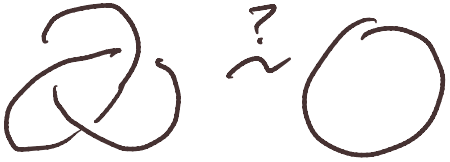
Answer. Yes.





i.e $K_1 \simeq T$

Exa 2



Invariant of knots

Φ ← an invariant of knots & links

$\{\text{links}\}/\sim \xrightarrow{\Phi} \mathcal{C}$ ← a set of groups, numbers, polynomials, ...

$\updownarrow$

$K_1 \sim K_2 \longmapsto \Phi(K_1) = \Phi(K_2)$

$\{\text{knot diagrams}\}/\text{RMs}$

Coloring invariant

D: an oriented diagram, $A(D)$: the set of arcs of D.



$A(D) = \{a_1, a_2, a_3\}$

A proper 3-coloring (q (prime number)) is a map

$$p: A(D) \to \{0, 1, 2\} \Rightarrow \mathbb{Z}_q$$

satisfying

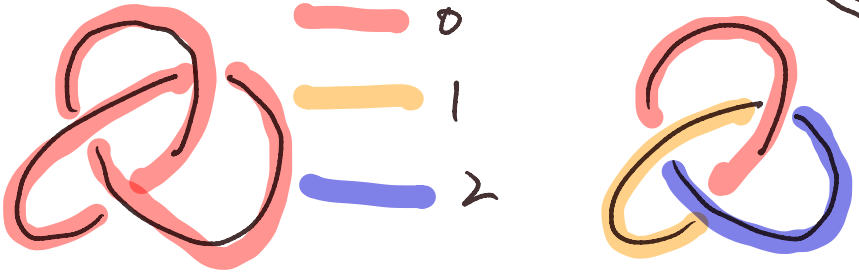


$p(a_i) = p(a_j) = p(a_k)$

or

$p(a_i) \neq p(a_j) \neq p(a_k)$

$\Rightarrow p(a_k) = 2p(a_j) - p(a_i) \mod q$



Trivial coloring

A (nontrivial) proper 3-coloring.

If D has a nontrivial proper 3-coloring, then D is called 3-colorable.

Exa



Not 3-colorable ✗

Three trivial proper 3-colorings.





✗

∃ non-trivial proper 3-coloring. i.e 3-colorable.



∄ non-trivial proper 3-coloring, i.e. Not 3-colorable.

???

Lemma 3-colorability is preserved under RM.

In other words, if $D \sim D'$ and $\exists P : A(D) \to \{0,1,2\}$, then

$\exists! P_{D'} : A(D') \to \{0,1,2\}$ corresponding to P_D

Idea of proof) Assume D' can be obtained from D by one of RM's

Case 1.

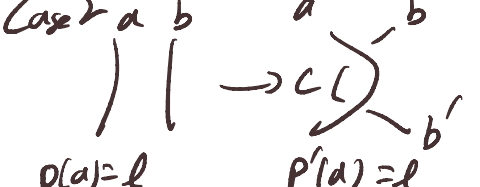


D $p(a)=l$



D' $p'(a)=l=p'(a')$

Case 2



$p(a)=l$
$p(b)=m$

$p'(a)=l$
$p'(b)=m=p'(b')$

If $l=m$, $p'(c)=l$

If $l \neq m$, $p'(c)=n$

□

Thm If $D \sim D'$, then

$$\#\{\text{proper } 3\text{-colorings of } D\} = \#\{\text{proper } 3\text{-colorings of } D'\}$$

$\Rightarrow$ 3-coloring invariant.

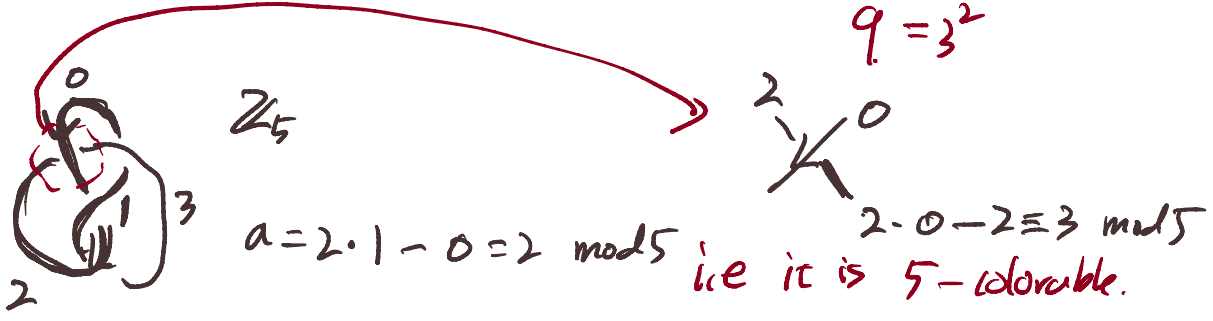
Exa



$\#\{\text{proper } 3\text{-colorings of } D\} = 3 \neq \#\{\text{proper } 3\text{-colorings of } T\} \geq 9.$

$\| \quad 9 = 3^2$

Exa



$a = 2 \cdot 1 - 0 = 2 \bmod 5$

$2 \cdot 0 - 2 \equiv 3 \bmod 5$

i.e. it is 5-colorable.

Remark 

$\#\{\text{proper 5-colorings of } D\} = 25 = 5^2$

Exer $\#\{\text{proper } q\text{-colorings of } D\} = q^k$.

Idea





$c = 2b - a$

$a - 2b + c = 0$.

$\Rightarrow$

$$\begin{array}{c} \\ c_1 \\ c_2 \\ c_3 \\ c_4 \end{array} \begin{array}{c} \begin{array}{cccc} a_1 & a_2 & a_3 & a_4 \end{array} \\ \begin{pmatrix} -2 & 1 & 1 & 0 \\ & & & \\ & & & \\ & & & \end{pmatrix} \end{array} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$\in M_{4\times 4}(\mathbb{Z}_5)$

$a_2 - 2a_1 + a_3 = 0$

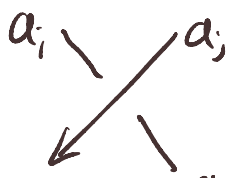
Since 5 is prime, $\mathbb{Z}_5$ is field.

So, we can apply linear algebra directly.

Exa 

3-colorable

Quandle and quandle coloring



Q: a set w/ a binary operation $*$.

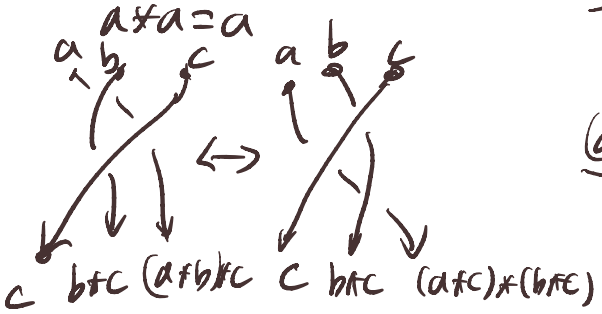
$P: A(D) \to Q$. $a_k = a_i * a_j$ $\forall$ c: crossing.

To obtain a colony invariant by $(Q, *)$, $(Q, *)$ should satisfy some conditions:



$\forall a, b \in Q$, $\exists! c$ s.t. $c * a = b$.

$(a*b)*c = (a*c)*(b*c)$



Def A quandle is a set Q w/ bin. oper. $*$ s.t

① $\forall a,\ a*a=a$

② $\forall a,b,\ \exists!\ c$ s.t $c*a=b$

③ $\forall a,b,c\ \ (a*b)*c=(a*c)*(b*c)$

A proper coloring of D by $(Q,*)$ is a map $A(D)\to Q$ s.t 

$c=a*b$

Thm If $D\sim D'$, then $\exists$ 1-1 correspondence between

$\{$proper coloring of D by $Q\} \longleftrightarrow \{$proper coloring of D by $Q\}$

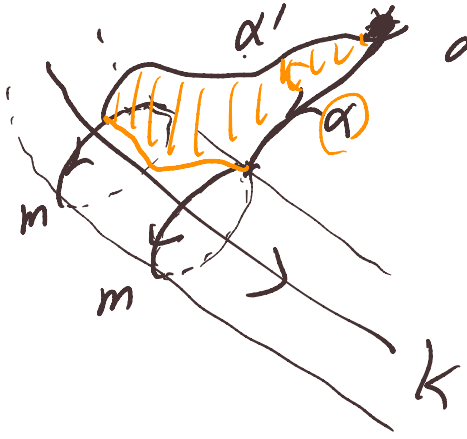
Exa. $(\mathbb{Z}_p,*)$ $a*b=2b-a$: a dihedral quandle.

- $(G,\cdot)$: a group, $\Rightarrow (G,*)$, $a*b=bab^{-1}$: a conjugation quandle.
- $(\mathbb{Z}[t^{\pm1}],*)$ $a*b=ta+(1-t)b$: Alexander quandle.

Knot quandle

tubular nbd of K

For a knot K, we consider $S^3 - N(K)$



$\alpha \cdot m \sim \alpha' \cdot m$ $\quad Q(K) = \{[\alpha \cdot m]\}$

Define $[\alpha \cdot m] * [\beta \cdot m]$

$= [\beta \cdot m^{-1} \cdot \beta^{-1} \cdot \alpha \cdot m]$.



$\Rightarrow (Q(K), *)$: a quandle.

Q knot quandle.

Lemma If $K \sim K'$, then $Q(K)$ and $Q(K')$ are isomorphic.



Remark



There is natural correspondance between $\pi_1(S^3 - N(K))$ and $Q(K)$

Thm Let K, K' be oriented knots.

If $Q(K) \cong Q(K')$, then $K \sim K'$ or $-K \sim K'$ or $K^* \sim K'$ or $-K^* \sim K'$.

where $-K$: ori. reversing

K^* : mirror image

So, knot quandle is called an almost complete invariant.